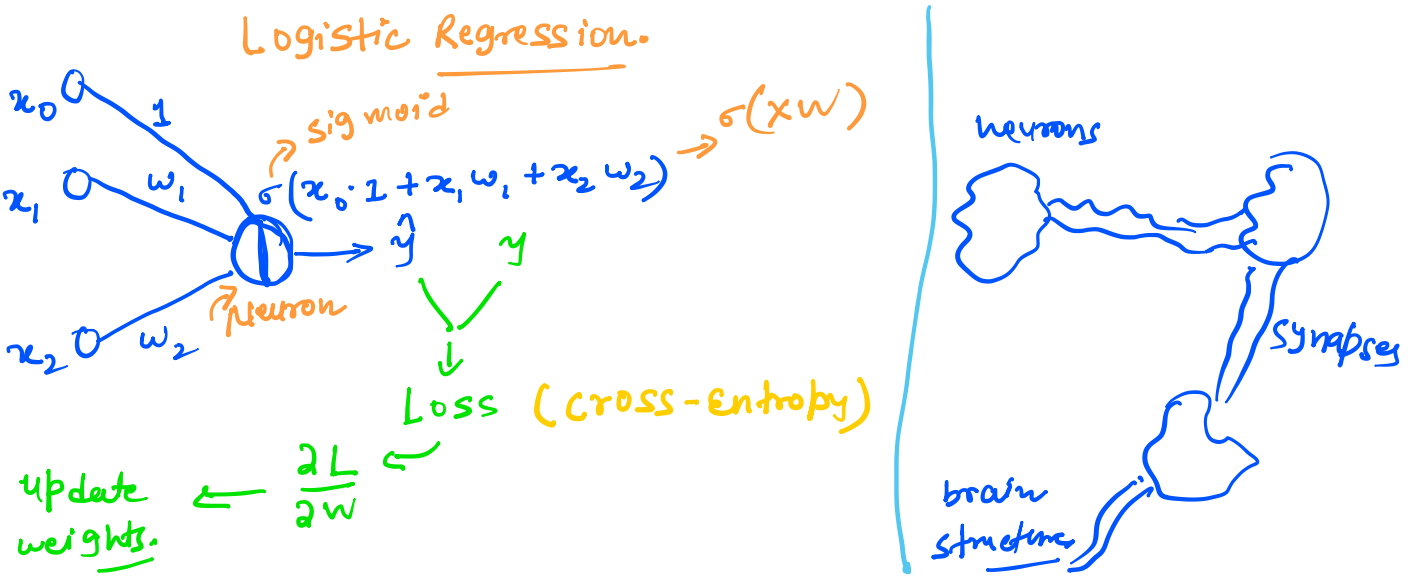
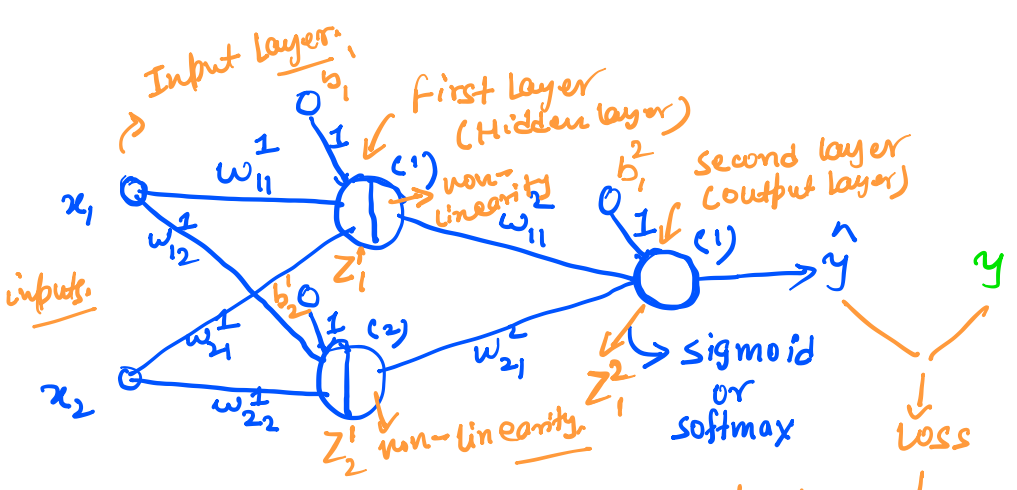
# Introduction to Neural Networks.

## How Do Several Neurons Learn Together?




1st layer weights

$w_{11}^1, w_{12}^1, w_{21}^1, w_{22}^1$

$\rightarrow W^1$

2nd layer weights

$w_{11}^2 \quad w_{21}^2$

$\rightarrow W^2$

parameters of the model $\rightarrow w_{11}^{(1)}, w_{12}^{(1)}, w_{21}^1, w_{22}^1, w_{11}^2, w_{21}^2$

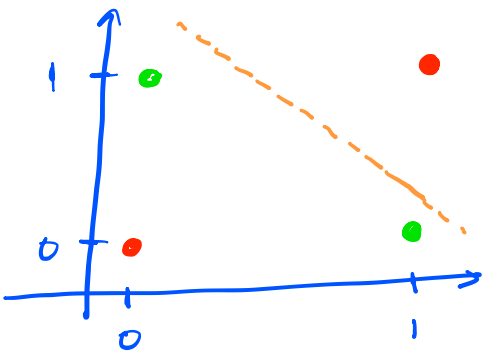
$\rightarrow$ 6 parameters.

$$z_1^1 = x_1 \cdot w_{11}^{(1)} + x_2 \cdot w_{21}^{(1)} + b_1^1$$

$$z_2^1 = x_1 \cdot w_{12}^{(1)} + x_2 \cdot w_{22}^{(2)} + b_2^1$$

## XOR gate

| $x_1$ | $x_2$ | $y$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |




$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times 1}$$

$$W^1 = \begin{bmatrix} w_{11}^1 & w_{21}^1 \\ w_{12}^1 & w_{22}^1 \end{bmatrix}_{2\times 2} \qquad b^1 = \begin{bmatrix} b_1^1 \\ b_2^1 \end{bmatrix}_{2\times 1}$$

$$Z^1 = \begin{bmatrix} z_1^1 \\ z_2^1 \end{bmatrix}_{2\times 1}$$

$$W^1 \cdot X = \begin{bmatrix} w_{11}^1 x_1 + w_{21}^1 x_2 \\ w_{12}^1 x_1 + w_{22}^1 x_2 \end{bmatrix}$$

$$W^1 X + b^1 = \begin{bmatrix} w_{11}^1 x_1 + w_{21}^1 x_2 + b_1 \\ w_{12}^1 x_1 + w_{22}^1 x_2 + b_2 \end{bmatrix}_{2\times 1} = Z^1$$

$$\boxed{Z^1 = W^1 X + b^1}$$

$$a^1 = \sigma(Z^1)$$

$$Z^2 = W^2 \cdot Z^1 + b^2$$

($Z^2$: $1\times 1$, $W^2$: $1\times 2$, $Z^1$: $2\times 1$, $b^2$: $1\times 1$)

$$W^2 = \begin{bmatrix} w_{11}^2 & w_{12}^2 \end{bmatrix}_{1\times 2}$$

$$b^2 = \begin{bmatrix} b_1^2 \end{bmatrix}_{1\times 1}$$

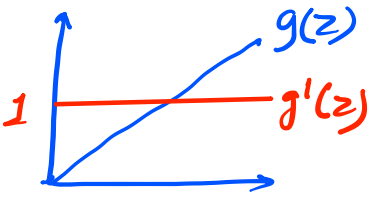
$$Z = \begin{bmatrix} z_1^1 \\ z_2^1 \end{bmatrix}_{2\times 1}$$

$$\hat{y} = \sigma(Z_1^2)$$

# Activations:

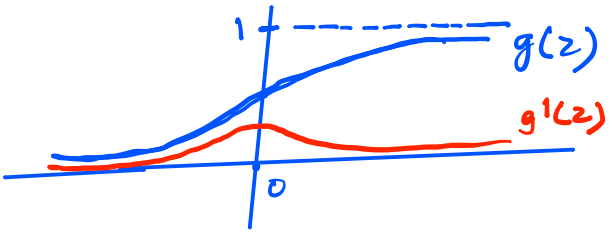
## ① Identity:

$$g(z) = z, \quad g'(z) = 1$$



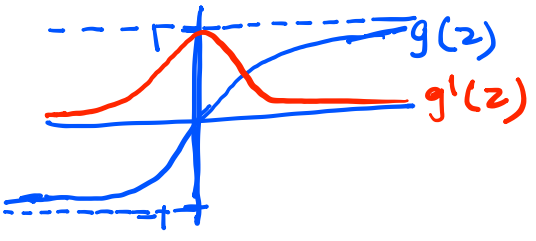

## ② Sigmoid:

$$g(z) = \frac{1}{1+e^{-z}}, \quad g'(z) = g(z)\cdot(1-g(z))$$



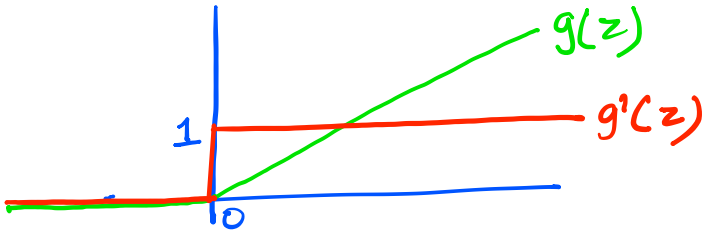

## ③ tanh:

$$g(z) = \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}, \quad g'(z) = 1-\tanh^2(z)$$




## ④ Rectified Linear Unit (ReLU):

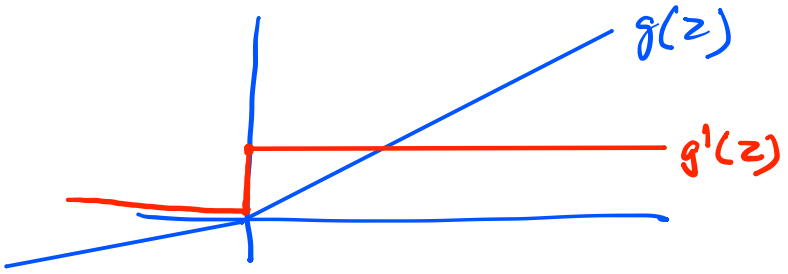
$$g(z) = \max(0, z), \quad g'(z) = \begin{cases} 1 & z > 0 \\ 0 & z \leq 0 \end{cases}$$




⑤ Leaky ReLU:

$$g(z) = \begin{cases} z, & z>0 \\ \alpha z & z<0 \end{cases} \qquad g'(z) = \begin{cases} 1 & z>0 \\ \alpha & z\leq 0 \end{cases}$$

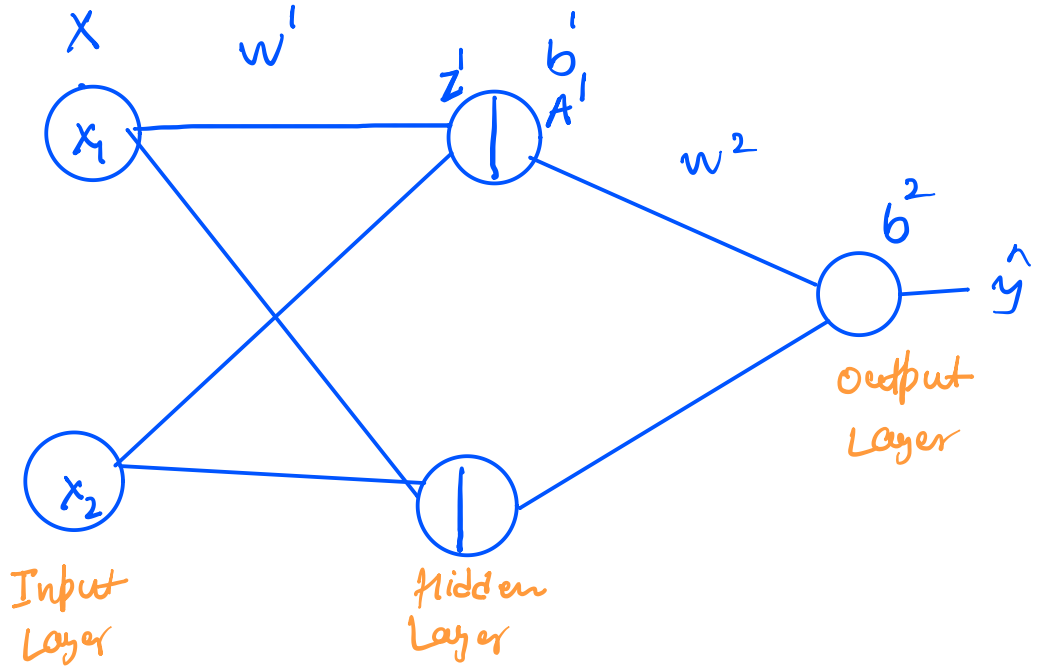
↳ generally 0.1, 0.2




multiple layers without activation fn:

$((X w^1) w^2) w^3 _____$

$(X P)$, $P = w^1 w^2 w^3 __$



$X = 1 \times 2 \longrightarrow m \times 2$

$W^1 = 2 \times 2$, $W^2 = 2 \times 1$

$Z^1 = X W^1 + b^1 = 1 \times 2 \rightarrow m \times 2$

$A^1 = \sigma(Z^1) = 1 \times 2 \rightarrow m \times 2$ (activation function)

$Z^2 = A^1 W^2 + b^2 = 1 \times 1 \rightarrow m \times 1$

$A^2 = \sigma(Z^2) = 1 \times 1 \longrightarrow m \times 1$

} forward pass.

Total no. of parameters =

$W^1 \rightarrow 4$, $W^2 \rightarrow 2$, $b^1 \rightarrow 2$, $b^2 \rightarrow 1$

$= 9$ parameters.

$\hat{y} = A^2$

**Loss function:** Binary cross-entropy.

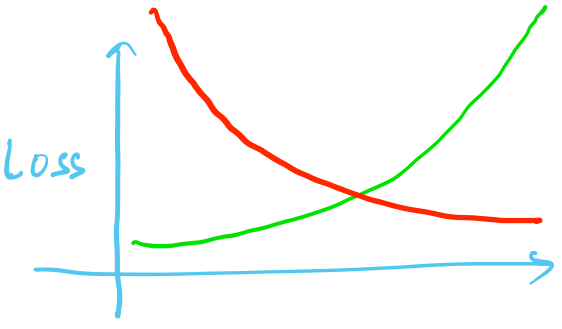
$$L(y, \hat{y}) = -[y \log \hat{y} - (1-y) \log(1-\hat{y})]$$

when $y = 1$,

$L(y, \hat{y}) = -\log \hat{y}$ (negative log likelihood)

when $y = 0$

$L(y, \hat{y}) = -\log(1-\hat{y})$




negative log likelihood loss (NLL loss)

$$J = \frac{1}{m}\sum_{i=1}^{m}\left[-y_i \log \hat{y}_i - (1-y_i)\log(1-\hat{y}_i)\right]$$

$$\frac{\partial J}{\partial w^2} = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial z^2} \times \frac{\partial z^2}{\partial w^2} \quad \text{(Chain rule).}$$

$$\frac{\partial J}{\partial w^1} = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial z^2} \times \frac{\partial z^2}{\partial A^1} \times \frac{\partial A^1}{\partial z^1} \times \frac{\partial z^1}{\partial w^1}$$

$$\frac{\partial J}{\partial A^2} = \frac{-y}{a} - (-1) \times \frac{1-y}{1-a} = \frac{-y}{a} + \frac{1-y}{1-a}$$

$y \quad \hat{y}_i$

$$\frac{\partial A^2}{\partial z^2} = a(1-a)$$

$$\frac{\partial J}{\partial z^2} = \left(\frac{-y}{a} + \frac{1-y}{1-a}\right) \cdot a(1-a)$$

$$\frac{\partial J}{\partial z^2} = a - y$$

$$z^2 = \sum_j a_j^1 w_j^2 + b^2$$

$$\frac{\partial z^2}{\partial w_j^2} = a_j^1$$

$$\frac{\partial J}{\partial w_j^2} = (a^2 - y) \cdot a_j^1$$

A derivative is a local sensitivity:

$$\frac{\partial J}{\partial w} = \frac{\text{small change in loss}}{\text{small change in } w}$$

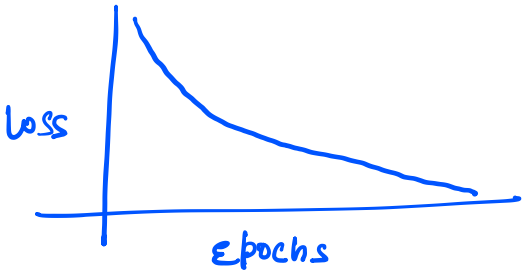
* **weight update** :

$$w^2 = w^2 - \eta \cdot \frac{\partial J}{\partial w^2} \rightarrow \text{vanilla optimizer}$$

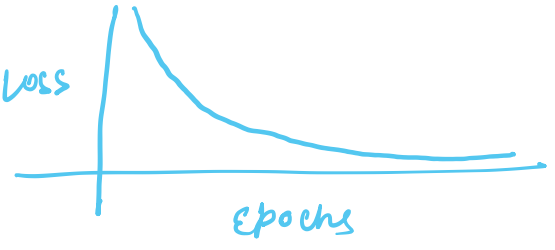
$$b^2 = b^2 - \eta \cdot \frac{\partial J}{\partial b^2}$$

$$w^1 = w^1 - \eta \cdot \frac{\partial J}{\partial w^1}$$

$$b^1 = b^1 - \eta \cdot \frac{\partial J}{\partial b^1}$$




increase the LR, or epochs




balanced LR.

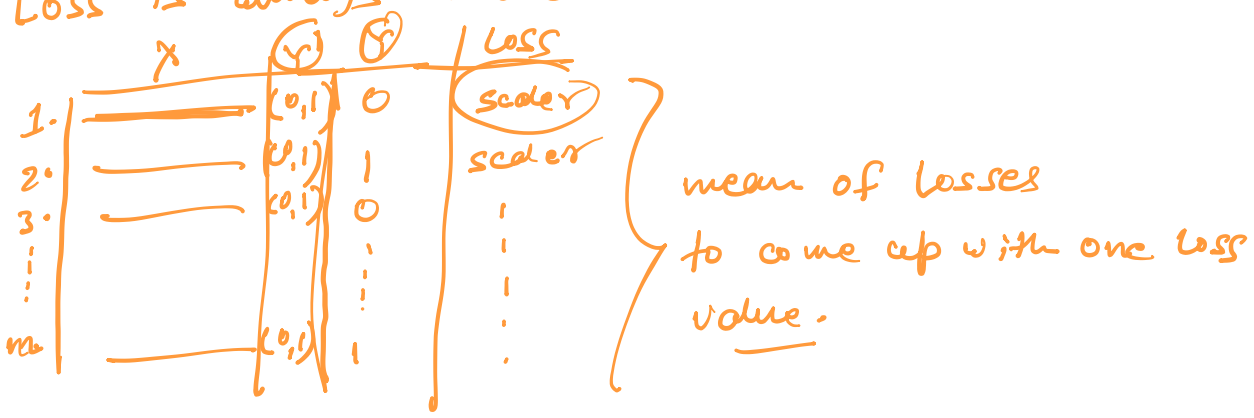
Diff. versions of gradient descent

(1) Stochastic gradient descent
↳ single sample chosen randomly (w/o replacement) is used to update the weights.

(2) full grad. descent: ⇒ all $m$ samples at once are used to update weights.

(3) batch grad. descent: $k << m$ samples at once, are used to update weights.

Loss is always scaler.




$W2 = 2 \times 1$

$dz2 = 1 \times 1$

$A1 = 1 \times 2$

$dw^2 = 2 \times 1$

$\frac{\partial l}{\partial w^2} = dz^2 \quad A1$

$1 \times 2 \quad 1 \times 1 \quad X$

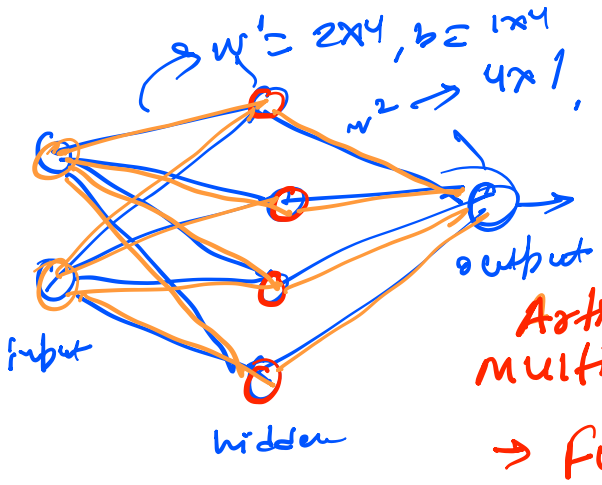
$2 \times 1 \quad 1 \times 1 = 2 \times 1 = W2$ shape.

$a = b + c \qquad a = b \times c \qquad a = \log(b)$

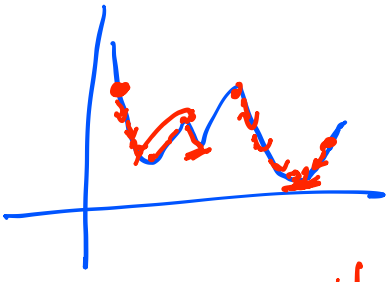
$\frac{\partial a}{\partial b} = 1 \qquad \frac{\partial a}{\partial b} = c \qquad \frac{\partial a}{\partial b} = \frac{1}{b}$

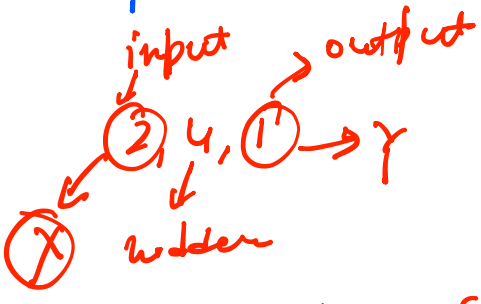
$a = \frac{b}{c} = b \times \frac{1}{c} \qquad \frac{1}{c} \times b$

$\frac{\partial a}{\partial c} = \frac{1}{c}, \quad \frac{\partial a}{\partial c} = ?$




Artificial neural network (ANN)
Multilayer perceptron (MLP)
→ fully connected network
(FC) network






hidden_layer_sizes = [4], [5, 3]

